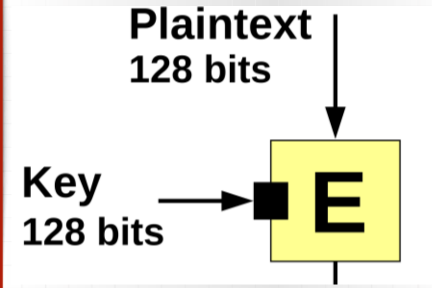
Cybersecurity
Symmetric Ciphers



Cybersecurity

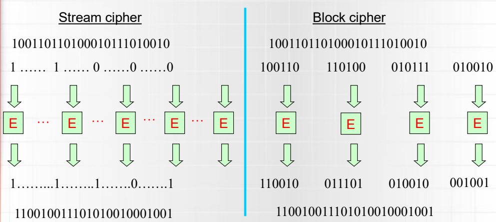
Agenda

- Shannon Principles
- Feistel Cipher
- DES Algorithm
- AES
- Modes of operation
- Stream Ciphers

Cybersecurity

Symetric Ciphers

- There are two classes: Block ciphers and steam ciphers



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Block Ciphers

- A ***block cipher*** is an encryption scheme which breaks up the plaintext message into blocks of a fixed length and produces ciphertext blocks of the same length.

- Block ciphers encrypt one block at a time, using a complex encryption function
- Examples
  - DES: operates on blocks of 64 bits
  - AES: operates on blocks of 128 bits
- Block ciphers can be used in various modes – ***modes of operation***

Cybersecurity

Substitution-Permutation Ciphers

- 1949 Shannon introduced idea of substitution-permutation (S-P) networks
  - modern substitution-transposition product cipher
- These form the basis of modern block ciphers
- S-P networks are based on the two primitive cryptographic operations :
  - *substitution* (S-box)
  - *permutation* (P-box) (transposition)
- Provide *confusion* and *diffusion* of message

Cybersecurity

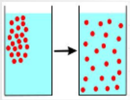
Diffusion and Confusion

- Introduced by Claude Shannon to prevent cryptanalysis based on statistical analysis
  - Assume the attacker has some knowledge of the statistical characteristics of the plaintext
  - Cipher needs to completely obscure statistical properties of original message

Cybersecurity

Diffusion and Confusion

- More practically Shannon suggested combining elements to obtain:
  - **Diffusion** – dissipates statistical structure of plaintext over bulk of ciphertext
  - **Confusion** – makes relationship between ciphertext and key as complex as possible



Cybersecurity

Diffusion and Confusion

- **<u>Diffusion</u> is used to dissipate the statistical structure of the plaintext into long range statistical properties of the cyphertext**
  - We try to make the statistical relationship between plaintext and cyphertext complex so they key cannot be derived- ideally by having <u>each plaintext bit affect as many as possible cyphertext bits</u>.
  - In cypher design, we try to get the cyphertext symbol, digraph and trigraph frequencies as evenly distributed as possible, and ideally flipping a bit of the plaintext <u>will result in a 50% probability of each bit</u> flipping in the cyphertext
  - Diffusion is usually achieved through <u>repeat application of a permutation</u> function
  - Sometimes seen as a 'P-Box' in cyphers
- **<u>Confusion</u> is used to make the relationship between the cyphertext and the key as difficult as possible**
  - Usually achieved through application of a <u>complex substitution function</u>
  - Usually seen in the form of a n x m bit 'S-box'
  - Think of a n-bit address line into a n x m-bit RAM (storing a non-linear function)

Cybersecurity

Feistel Cipher Structure

- Horst Feistel developed the **feistel cipher**
  - implements Shannon's substitution-permutation network concept
  - partitions input block into two halves
  - process through multiple rounds
  - perform a substitution on left data half
  - based on round function of right half & subkey
  - permutation swapping halves

Cybersecurity

Feistel Cipher Structure

Encryption:
- Plaintext = (L_0,R_0)
- For each round $i=1,2,\ldots n\text{-}1,n$, compute

  $L_i = R_{i-1}$

  $R_i = L_{i-1} \oplus F(R_{i-1},K_i)$

  where F is round function and K_i is subkey

Decryption:

Formula "works" for any function F
But only secure for certain functions F

- Ciphertext = (L_n,R_n)
- For each round $i=n,n-1,\ldots,1$, compute

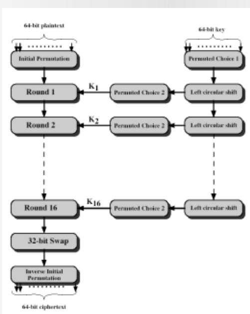
  $R_{i-1} = L_i$

  $L_{i-1} = R_i \oplus F(R_{i-1},K_i)$

  where F is round function and K_i is subkey
- Plaintext = (L_0,R_0)

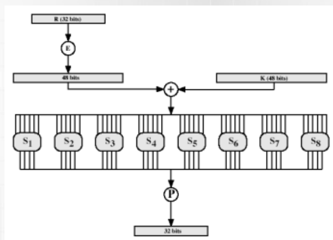
Cybersecurity

DES – Encryption



Cybersecurity

DES – Round Function F(R,K)



Cybersecurity

DES – Substitution Boxes

S1
14, 4, 13, 1, 2, 15, 11, 8, 3, 10, 6, 12, 5, 9, 0, 7,
0, 15, 7, 4, 14, 2, 13, 1, 10, 6, 12, 11, 9, 5, 3, 8,
4, 1, 14, 8, 13, 6, 2, 11, 15, 12, 9, 7, 3, 10, 5, 0,
15, 12, 8, 2, 4, 9, 1, 7, 5, 11, 3, 14, 10, 0, 6, 13,

- 8 S-boxes
- Each S-Box maps 6 to 4 bits
  - outer bits 1 & 6 (**row** bits) select the row
  - inner bits 2-5 (**col** bits) select the column
  - For example, in S1, for input 011001,
    - the row is 01 (row 1)
    - the column is 1100 (column 12).
    - The value in row 1, column 12 is 9
    - The output is 1001.
- result is 8 X 4 bits, or 32 bits

Cybersecurity

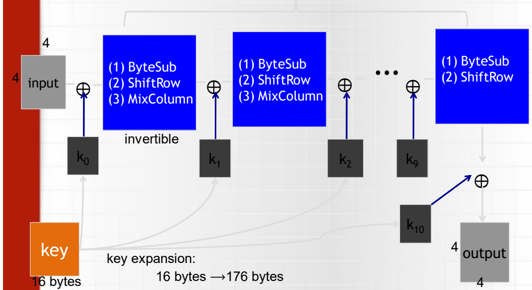
AES

- 1997: NIST publishes request for proposal
- 1998: 15 submissions.
- 1999: NIST chooses 5 finalists
- 2000: NIST chooses Rijndael as AES (designed in Belgium)

Key sizes: 128, 192, 256 bits. Block size: 128 bits

Cybersecurity

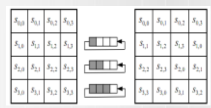
AES 128



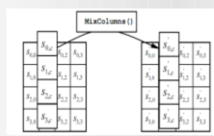
Cybersecurity

AES – Round Function

- **ByteSub**: a 1 byte S-box. 256 byte table (easily computable)
- **ShiftRows**:



- **MixColumns**:



Cybersecurity

AES – Byte Substitution

- Simple substitution of each byte
- Uses one table of 16x16 bytes containing a permutation of all 256 8-bit values
- Each byte of state is replaced by byte in row (left 4-bits) & column (right 4-bits)
  - eg. byte {95} is replaced by row 9 col 5 byte
  - which is the value {2A}
- S-box is constructed using a defined transformation of the values in GF(2^8)
- Designed to be resistant to all known attacks

Cybersecurity

AES – Shift Rows

- Curucular byte shift in each each
  - 1^{st} row is unchanged
  - 2^{nd} row does 1 byte circular shift to left
  - 3rd row does 2 byte circular shift to left
  - 4th row does 3 byte circular shift to left
- Decrypt does shifts to right
- Since state is processed by columns, this step permutes bytes between the columns

Cybersecurity

AES – Mix Columns

- Each column is processed separately
- Each byte is replaced by a value dependent on all 4 bytes in the column
- Effectively a matrix multiplication in GF(2^8) using prime poly m(x) =$x^8+x^4+x^3+x+1$

$$\begin{bmatrix} 02 & 03 & 01 & 01 \\ 01 & 02 & 03 & 01 \\ 01 & 01 & 02 & 03 \\ 03 & 01 & 01 & 02 \end{bmatrix} \begin{bmatrix} s_{0,0} & s_{0,1} & s_{0,2} & s_{0,3} \\ s_{1,0} & s_{1,1} & s_{1,2} & s_{1,3} \\ s_{2,0} & s_{2,1} & s_{2,2} & s_{2,3} \\ s_{3,0} & s_{3,1} & s_{3,2} & s_{3,3} \end{bmatrix} = \begin{bmatrix} s'_{0,0} & s'_{0,1} & s'_{0,2} & s'_{0,3} \\ s'_{1,0} & s'_{1,1} & s'_{1,2} & s'_{1,3} \\ s'_{2,0} & s'_{2,1} & s'_{2,2} & s'_{2,3} \\ s'_{3,0} & s'_{3,1} & s'_{3,2} & s'_{3,3} \end{bmatrix}$$

Cybersecurity

AES – Round Key

- XOR state with 128-bits of the round key
- Again processed by column (though effectively a series of byte operations)
- Inverse for decryption is identical since XOR is own inverse, just with correct round key
- Designed to be as simple as possible

Cybersecurity

Properties of block ciphers

- Block ciphers do propagate errors (to a limited extent), but are quite flexible and can be used in different ways in order to provide different security properties (in some cases to achieve some of the benefits of stream ciphers).
- The properties of cryptographic algorithms are not only affected by algorithm design, but also by the ways in which the algorithms are used.
- Different modes of operation can significantly change the properties of a block cipher.

Cybersecurity

Properties of block ciphers

- Modern version of a **codebook cipher**
- In effect, a block cipher algorithm yields a huge number of codebooks
  - Specific codebook determined by key
- It is OK to use same key for a while
  - Just like classic codebook
- Change the key, get a new codebook

Cybersecurity

Multiple Blocks

- How to encrypt multiple blocks?
- A new key for each block?
  - As bad as (or worse than) a one-time pad!
- Encrypt each block independently?
- Make encryption depend on previous block(s), i.e., “chain” the blocks together?
- How to handle partial blocks?

Cybersecurity

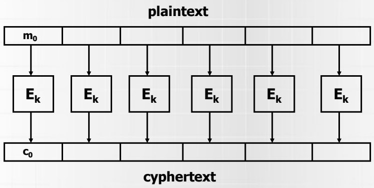
Block Cipher Modes

- We discuss 4 (many others)
- Electronic Codebook (**ECB**) mode
  - Encrypt each block independently
  - There is a serious weakness
- Cipher Block Chaining (**CBC**) mode
  - Chain the blocks together
  - Better than ECB, virtually no extra work
- Output Feedback (**OFM**) mode
- Counter Mode (**CTR**) mode
  - Like a stream cipher (random access)

Cybersecurity

Electronic Code Book (ECB)

– **Each 64 bit block is encrypted separately.**



Cybersecurity

Electronic CodeBook (ECB):

- Notation: $C=E(P,K)$
- Given plaintext $P_0,P_1,\ldots,P_m,\ldots$
- Obvious way to use a block cipher is

Encrypt	Decrypt
$C_0 = E(P_0, K)$,	$P_0 = D(C_0, K)$,
$C_1 = E(P_1, K)$,	$P_1 = D(C_1, K)$,
$C_2 = E(P_2, K),\ldots$	$P_2 = D(C_2, K),\ldots$

- For a fixed key K, this is an electronic version of a codebook cipher
- A new codebook for each key
- Example

Cybersecurity

ECB Cut and Paste Attack

- Suppose plaintext is
  `Alice digs Bob. Trudy digs Tom.`
- Assuming 64-bit blocks and 8-bit ASCII:
  P_0 = "`Alice di`", P_1 = "`gs Bob. `",
  P_2 = "`Trudy di`", P_3 = "`gs Tom. `"
- Ciphertext: C_0,C_1,C_2,C_3
- Trudy cuts and pastes: C_0,C_3,C_2,C_1
- Decrypts as
  `Alice digs Tom. Trudy digs Bob.`

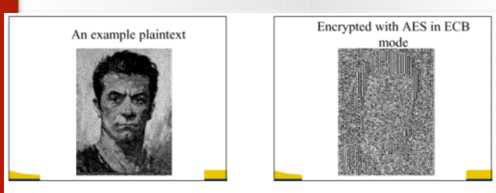
Cybersecurity

ECB Weakness

- Suppose $P_i = P_j$
- Then $C_i = C_j$ and Trudy knows $P_i = P_j$
- This gives Trudy some information, even if she does not know P_i or P_j
- Trudy might know P_i
- Is this a serious issue?

Cybersecurity

ECB In pictures



Cybersecurity

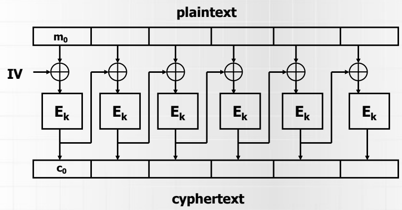
Electronic CodeBook (ECB):

- Identical plaintext blocks (under the same key) result in identical ciphertext.
- Vulnerable to dictionary attacks
- Chaining dependency: blocks are enciphered independently of other blocks.
- Error propagation: one or more bit errors in a single ciphertext affect decipherment of that block only.
- ECB is not recommended for messages longer than one block, or if keys are reused for more than one-block message.
- Security of ECB may be improved by inclusion of random padding bits in each block.

Cybersecurity

Cypher Block Chaining (CBC)

- Blocks are chained together
- IV is some predetermined value



Cybersecurity

Cypher Block Chaining (CBC)

- Identical plaintexts: identical ciphertext blocks result when the same plaintext is enciphered under the key and *IV*.
- Chaining dependency: a ciphertext c_j depends on x_j and all preceding plaintext blocks ⇒ rearranging the order of ciphertext blocks affects decryption.
- Error propagation: a single bit error in ciphertext block c_j affects decipherment of c_j and c_{j+1}.
- Error recovery: CBC is *self-synchronizing* in the sense that if an error occurs in block c_j, c_{j+2} is correctly recovered.
- *IV* is not secret but needs integrity.

Cybersecurity

Cypher Block Chaining (CBC)

- Blocks are "chained" together
- A random initialization vector, or IV, is required to initialize CBC mode
- IV is random, but need not be secret

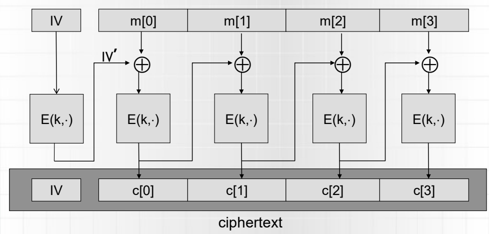
Encryption	**Decryption**
$C_0 = E(IV \oplus P_0, K)$,	$P_0 = IV \oplus D(C_0, K)$,
$C_1 = E(C_0 \oplus P_1, K)$,	$P_1 = C_0 \oplus D(C_1, K)$,
$C_2 = E(C_1 \oplus P_2, K)$,…	$P_2 = C_1 \oplus D(C_2, K)$,…

- Example

Cybersecurity

Cypher Block Chaining (CBC)

- CBC with Unique Ivs
- unique IV means: (k,IV) pair is used for only one message
- may be predictable so use E(k,·) as PRF



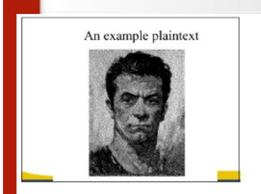
Cybersecurity

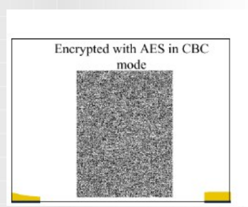
Cypher Block Chaining (CBC)

- Identical plaintext blocks yield different ciphertext blocks
- Cut and paste is still possible, but more complex (and will cause garbles)
- If C_1 is garbled to, say, G then
  $P_1 \neq C_0 \oplus D(G, K)$, $P_2 \neq G \oplus D(C_2, K)$
- But $P_3 = C_2 \oplus D(C_3, K)$, $P_4 = C_3 \oplus D(C_4, K)$,…
- Automatically recovers from errors!

Cybersecurity

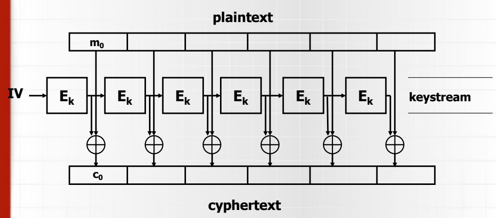
Cypher Block Chaining (CBC)





Cybersecurity

Output Feedback Mode (OFM)



Cybersecurity

Output Feedback Mode (OFM)

- Identical plaintexts result in identical cyphertexts when the same plaintext is enciphered using the same key and IV.
- Chaining dependencies:
  The keystream is plaintext independent
- Error propagation:
  one or more bit errors in any cyphertext block results only in decipherment of that block in the precise position of error
- Error recovery:
  OFB recovers from cyphertext bit errors but not bit loss (results in unalignment of keystream)
- Throughput:
- Keystream may be independently calculated (e.g. precomputed)
- IV must be changed if the key is reused

Cybersecurity

Counter Mode (CTR)

- CTR is popular for random access
- Use block cipher like stream cipher

Encryption	**Decryption**
$C_0 = P_0 \oplus E(IV, K)$,	$P_0 = C_0 \oplus E(IV, K)$,
$C_1 = P_1 \oplus E(IV{+}1, K)$,	$P_1 = C_1 \oplus E(IV{+}1, K)$,
$C_2 = P_2 \oplus E(IV{+}2, K),\ldots$	$P_2 = C_2 \oplus E(IV{+}2, K),\ldots$

- CBC can also be used for random access!!!
- Example